

# Sequential ETL Pipeline — Tasks & Dependencies

## Simple\_etl.py

```
from airflow.sdk import dag, task
from datetime import datetime, timedelta

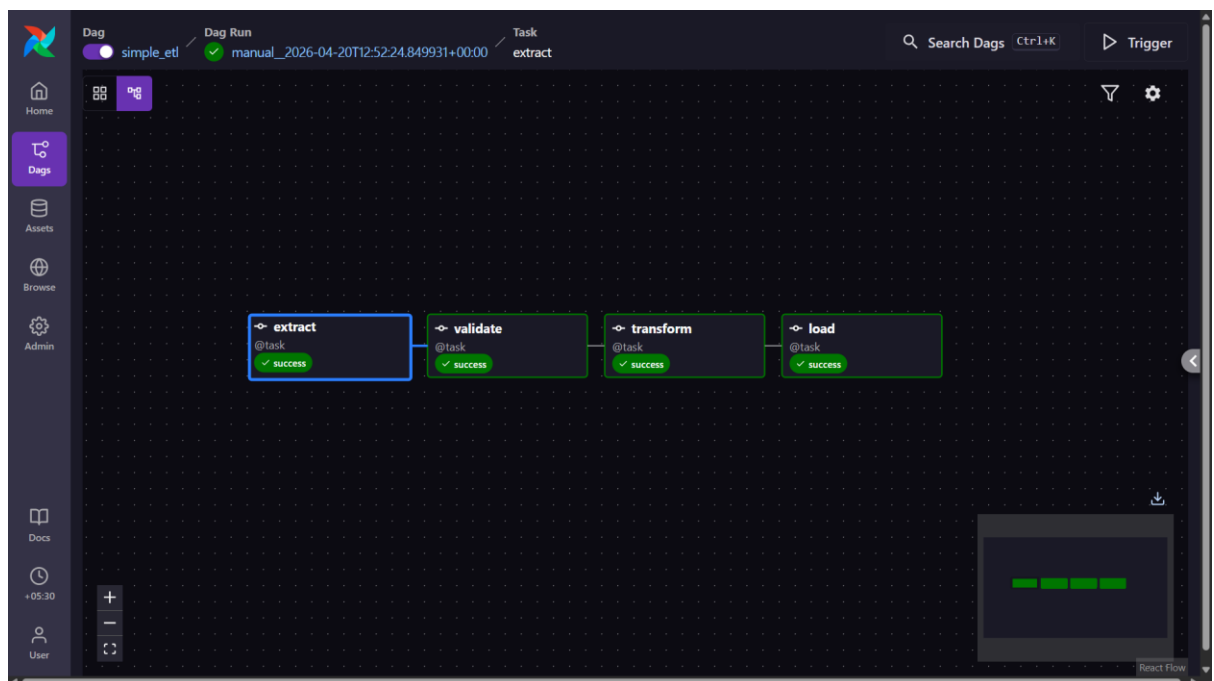
@dag(
    dag_id="simple_etl",
    schedule="@daily",
    catchup=False,
    tags=["Assignment-1", "simple_etl", "Uday"]
)
def simple_etl():
    @task()
    def extract(**context):
        products=[
            {"product_id": 1, "name": "Product A", "price": 10.0},
            {"product_id": 2, "name": "Product B", "price": 20.0},
            {"product_id": 3, "name": "Product C", "price": 30.0},
            {"product_id": 4, "name": "Product D", "price": 40.0},
        ]
        print(f"Extracting sales data for date: {context['ds']}")
        for i in products:
            print(f"Product ID: {i['product_id']}, Name: {i['name']}, Price: {i['price']}")
        return products
    @task()
    def validate(products):
        print("Validating data...")
        print(f"Validation passed – {len(products)} records found")
        return products
    @task()
    def transform(products):
        print("Transforming data...")
        print("Applying 10% discount to all records")
        print("Transformed data:")
        for i in products:
            i['price'] = i['price'] * 0.9
            print(f"Product ID: {i['product_id']}, Name: {i['name']}, Discounted Price: {i['price']}")
        return products
    @task()
    def load(products, **context):
        print("Loading to warehouse...")
```

```

        print(f"Successfully loaded {len(products)} records on
{context['ds']}")
        for i in products:
            print(f"Product ID: {i['product_id']}, Name: {i['name']}, Final
Price: {i['price']}")
            data = extract()
            validated = validate(data)
            transformed = transform(validated)
            load(transformed)
simple_etl = simple_etl()

```

1. Graph View — all 4 tasks GREEN in correct left-to-right order:  
extract → validate → transform → load



2. Task Log of transform — showing each product printed with its discounted price

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main area shows a DAG named 'simple\_etl' with a task named 'transform' highlighted. The task status is 'Success'. The task log is open, showing the following details:

| Operator | Start Date          | End Date            | Duration     | Dag Version |
|----------|---------------------|---------------------|--------------|-------------|
| @task    | 2026-04-21 15:24:01 | 2026-04-21 15:24:01 | 00:00:00.188 | v2          |

The log messages are as follows:

```
Log message source details sources=["/usr/local/airflow/logs/dag_id=simple_etl/run_id=manual_2026-04-21T09:53:59.147318+00:00/task_id=transform"]
2 [2026-04-21 15:24:01] WARNING - Astro managed secrets backend is disabled
3 [2026-04-21 15:24:01] INFO - DAG bundles loaded: dags-folder
4 [2026-04-21 15:24:01] INFO - Filling up the DagBag from /usr/local/airflow/dags/simple_etl.py
5 [2026-04-21 15:24:01] INFO - TaskInstance Details: dag_id=simple_etl task_id=transform dagrun_id=manual_2026-04-21T09:53:59.147318+00:00
6 [2026-04-21 15:24:01] INFO - Done. Returned value was: [{"name": "Product A", "price": 9.0, "product_id": 1}, {"name": "Product B", "price": 18.0, "product_id": 2}, {"name": "Product C", "price": 27.0, "product_id": 3}, {"name": "Product D", "price": 36.0, "product_id": 4}]
7 [2026-04-21 15:24:01] INFO - Pushing xcom ti=RuntimeTaskInstance(id=8UID('019daf75-ff78-740a-bab2-4e75d8ebefaa'), task_id='transform', dag_id='simple_etl', run_id='manual_2026-04-21T09:53:59.147318+00:00')
8 [2026-04-21 15:24:01] INFO - Transforming data...
9 [2026-04-21 15:24:01] INFO - Applying 10% discount to all records
10 [2026-04-21 15:24:01] INFO - Transformed data:
11 [2026-04-21 15:24:01] INFO - Product ID: 1, Name: Product A, Discounted Price: 9.0
12 [2026-04-21 15:24:01] INFO - Product ID: 2, Name: Product B, Discounted Price: 18.0
13 [2026-04-21 15:24:01] INFO - Product ID: 3, Name: Product C, Discounted Price: 27.0
14 [2026-04-21 15:24:01] INFO - Product ID: 4, Name: Product D, Discounted Price: 36.0
```

### 3. Task Log of load — showing the "Successfully loaded" message with date and record count

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main area shows a DAG named 'simple\_etl' with a task named 'load' highlighted. The task status is 'Success'. The task log is open, showing the following details:

| Operator | Start Date          | End Date            | Duration     | Dag Version |
|----------|---------------------|---------------------|--------------|-------------|
| @task    | 2026-04-21 15:24:02 | 2026-04-21 15:24:02 | 00:00:00.157 | v2          |

The log messages are as follows:

```
Log message source details sources=["/usr/local/airflow/logs/dag_id=simple_etl/run_id=manual_2026-04-21T09:53:59.147318+00:00/task_id=load/att"]
2 [2026-04-21 15:24:02] WARNING - Astro managed secrets backend is disabled
3 [2026-04-21 15:24:02] INFO - DAG bundles loaded: dags-folder
4 [2026-04-21 15:24:02] INFO - Filling up the DagBag from /usr/local/airflow/dags/simple_etl.py
5 [2026-04-21 15:24:02] INFO - TaskInstance Details: dag_id=simple_etl task_id=load dagrun_id=manual_2026-04-21T09:53:59.147318+00:00 map_index=0
6 [2026-04-21 15:24:02] INFO - Loading to warehouse...
7 [2026-04-21 15:24:02] INFO - Successfully loaded 4 records on 2026-04-21
8 [2026-04-21 15:24:02] INFO - Product ID: 1, Name: Product A, Final Price: 9.0
9 [2026-04-21 15:24:02] INFO - Product ID: 2, Name: Product B, Final Price: 18.0
10 [2026-04-21 15:24:02] INFO - Product ID: 3, Name: Product C, Final Price: 27.0
11 [2026-04-21 15:24:02] INFO - Product ID: 4, Name: Product D, Final Price: 36.0
12 [2026-04-21 15:24:02] INFO - Done. Returned value was: None
```